

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
Kent, OH (open to NYC / SF relocation)

PROFESSIONAL SUMMARY

Applied AI engineer with production ML/AI depth - RAG systems, agentic LLM workflows, and FastAPI/Docker packaging - combining engineering discipline with customer-facing delivery to help Modal users get the most from GPU infrastructure. Master's degree (Kent State University). Track record of measurable ML/AI outcomes: ~35% retrieval precision improvement, >30% hallucination reduction, ~30% defect reduction in containerized ML services. Background spanning both engineering delivery and stakeholder-facing AI consulting at Progress Solutions, with comfort owning the full technical lifecycle from data to API to production. Python-native serverless GPU infrastructure aligns naturally with FastAPI/Docker production experience - Modal is the evolution of that stack.

CORE COMPETENCIES

Production ML/AI Systems • RAG & Retrieval Systems • Agentic LLM Workflows • FastAPI / Flask • Docker (Containerized ML Services) • PyTorch • Python • Customer-Facing Technical Delivery • Debugging ML Pipelines • Observability & Logging • CI/CD (Jenkins) • Stakeholder Communication

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS Healthcare | USA

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~35% and reduced irrelevant retrieval by >30%.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall by 15-20% on high-risk categories while holding precision >90%.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Client-facing healthcare AI consulting: HIPAA-conscious data governance, stakeholder-facing system-limitation docs, and communication of model behavior to non-technical healthcare administrators.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP | Hyderabad, India

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins**; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed **Grafana performance dashboards** tracking long-running queries, deadlocks, and I/O contention; cut incident recurrence ~25% and improved root-cause resolution speed ~30%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics | Hyderabad, India

- Built a **transformer-based video summarization** system across 5,000+ sessions; reduced manual review time by 60-70%.
- Implemented **document Q&A with RAG**; reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.
- Deployed as a **Flask-based API** with interface for non-technical staff - full end-to-end ownership of ML system in production.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; approximate-nearest-neighbor similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Agentic Pixel Character Synthesis & Animation Engine ongoing

Phased generative AI system: Stable Diffusion fine-tuning + LoRA, ControlNet + OpenPose/MediaPipe for pose conditioning, and an **LLM-based agent orchestrator**. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker - a Python-native containerized ML pipeline analogous to Modal's GPU serverless model.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped · Chrome Web Store

Full-stack browser extension built independently - Manifest V3, service workers. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider payload handling, seven-day translation cache, configs for 29 sites. Production-grade multi-provider AI integration shipped and maintained solo.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar, Replicate image models). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success rate **~25-30%**.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, LLMops, embeddings, vector search

ML Libraries: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, Stable Diffusion, video analytics

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging, cloud-ready deployment

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required